

* Ordinary Differential Equation

Topics:

- ① Separable variable Diff. Eqⁿ solve
- ② Linear 1st Order Diff. Eqⁿ solve + Application
- ③ Exact/Non-exact Diff. Eqⁿ solve
- ④ Homogeneous/Non-homogeneous Diff. Eqⁿ solve

Introduction to Differential Equation

A differential equation or simply D.E is any equation which contains derivatives, either ordinary or partial derivatives.

ODE ← ① Ordinary DE : $\frac{dy}{dx} + P(x)y = Q(x)$ ✓

PDE ← ② Partial DE : $\frac{\partial u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

$$\left(\frac{y}{x}\right)^2 = \left(\frac{y}{x}\right)^2 + 1 \Leftrightarrow \frac{y}{x} = \sqrt{\left(\frac{y}{x}\right)^2 + 1}$$

$$0 = y^2 + \frac{y^2}{x^2} = y^2 \left(1 + \frac{1}{x^2}\right)$$

Order: 3
Degree: 3

Order: 2

Degree: 1

Order of a DE:

The order of the differential equation (DE) is the largest derivatives present in the DE.

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

2nd

Order DE

Degree of DE:

This is the degree of the order of DE

$$\left(\frac{d^2y}{dx^2}\right)^1 + \frac{dy}{dx} + y = 0$$

e.g., $x^2 + 2x + 5 = 0$

→ 2nd Degree (দ্বিঘাত সমীকরণ)

→ Order : 2nd Order
→ Degree : 1st Degree

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = y \frac{d^3y}{dx^3} \Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = y^2 \left(\frac{d^3y}{dx^3}\right)^2$$

→ Order : 3rd
Degree : 2nd

$$\left(\frac{dy}{dx}\right)^4 + \frac{d^2y}{dx^2} + 5y = 0$$

Order : 2

Degree : 1 (not 4 ; as it's not highest order)

→ Classification of ODE:

① Classification by order

② Classification by linearity

- ① Linear DE
- ② Non-linear DE

③ Classification by homogeneity

- ① Homogeneous DE
- ② Non-homogeneous DE

→ Non-linear DE:

DV-Dependent Variable

Case ① DV × DV

Ex: $y^2 \frac{dy}{dx} + y = 0$

→ 1st order / 1st Degree / Non-linear DE

Case ② DV × DDV

Ex: $y \frac{dy}{dx} + y = 0$

→ Non-linear DE

Case ③ DDV × DDV

Ex: $\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} + y = 0$

DDV × DDV × DDV

Case ④ Transcendental function of DV

DDV- Derivative of Dependent Variable

Ex: $\frac{d^2y}{dx^2} + \frac{dy}{dx} + \sin y = 0$

Transcendental function

$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^y$

Transcendental function

$\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \ln y$

Note: ଏହା ଏକ Linear function

• n^{th} order ODE of the form:

$$G_n(x) \frac{d^n y}{dx^n} + G_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + G_1(x) \frac{dy}{dx} + G_0(x)y = f(x).$$

this DE is said to be -

① Homogenous if $f(x) = 0$

② Non-homogenous if $f(x) \neq 0$

Ex: ① $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = e^x$

$\underline{f(x) \neq 0}$

$\Rightarrow$ Non-homogenous

2nd order, 1st degree,
linear DE

② $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + 5y = 0$

$\Rightarrow$ Homogenous, 2nd order,
1st Degree, linear DE

Notes: 2nd order linear function

CT: Classify the eqⁿ(s) by writing L/NL/H/NH:

① $\ddot{y} + \sin y = 0 \rightarrow \text{NL, H}$

② $y''' + 3y' + 2y = x^2 \rightarrow \text{L, NH}$

③ $y''' + y'' + y = e^y \rightarrow \text{NL, H}$

④ $(1-y) \frac{dy}{dx} + y = x$
 $\Rightarrow \frac{dy}{dx} - y \cdot \frac{dy}{dx} + y = x \rightarrow \text{NL, NH}$

⑤ $(1-x) \frac{dy}{dx} + y = e^x$
 $\Rightarrow \frac{dy}{dx} - x \cdot \frac{dy}{dx} + y = e^x \rightarrow \text{L, NH}$

• Careful about x & y

$x + y = (1-y)/x - y \Leftrightarrow$

$\frac{dy}{dx} = \frac{y}{1-y} \quad \int \frac{1-y}{y} dy = \int \frac{1}{y} dy - \int 1 dy$

Ex-1: $\frac{dy}{dx} = \frac{y}{1-y}$

$\int \frac{1-y}{y} dy = \int \frac{1}{y} dy - \int 1 dy$

$\ln|y| - y = \ln|C|$

$\ln|y| - y = \ln|C| \Rightarrow y = C e^y$

→ Separable Equation:

A first order differential equation of the form:

$$\frac{dy}{dx} = f(x)g(y)$$

is said to be separable DE.

To solve this eqⁿ:-

• Method of Separation of variable:

$$\frac{dy}{dx} = f(x)g(y)$$

$$\Rightarrow \frac{dy}{g(y)} = f(x) dx$$

$$\Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx$$

$$\Rightarrow y = \dots \text{ (Ans.)}$$

Ex-1: $y \frac{dy}{dx} = 2x(y+1)$

$$\Rightarrow \frac{y \cdot dy}{y+1} = \int 2x \cdot dx$$

$$\Rightarrow \int \frac{y+1-1}{y+1} dy = \int 2x dx$$

$$\Rightarrow \int 1 dy - \int \frac{1}{y+1} dy = \int 2x dx$$

$$\Rightarrow y - \ln|y+1| = x^2 + c$$

(Ans.)

Diff. Eqⁿ → Anton X
Zill ✓

HW → Zill

Ex: 2.2 → (1-18), 23-26, 31, 32

Application of 1st order ODE

Ex-1: A culture initially has P_0 number of bacteria.

At $t=1$ hr, the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t ; determine the time necessary for the number of bacteria to triple.

Solⁿ:

$$\text{@ } t=0 \text{ hr, } P=P_0 \quad \left| \quad P(?)=3P_0\right.$$

$$\text{@ } t=1 \text{ hr, } P=\frac{3}{2}P_0$$

$$\rightarrow P(0)=P_0$$

$$\rightarrow P(1)=\frac{3}{2}P_0$$

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow \frac{dP}{dt} = KP \quad [K \rightarrow \text{const.}]$$

$$\Rightarrow \int \frac{dP}{P} = K \int dt$$

$$\Rightarrow \ln P - Kt = \ln C$$

$$\cdot \ln P - \ln C = Kt$$

$$\Rightarrow \ln\left(\frac{P}{C}\right) = Kt$$

$$\Rightarrow \frac{P}{C} = e^{Kt}$$

$$\Rightarrow P = Ce^{Kt}$$

$$\text{At } t=0, P=P_0$$

$$\textcircled{1} \Rightarrow P_0 = Ce^{K(0)} = C$$

$$\text{At } t=1, P=\frac{3}{2}P_0$$

$$\textcircled{1} \Rightarrow \frac{3}{2}P_0 = Ce^{K(1)} = P_0 e^K$$

$$\Rightarrow e^K = \frac{3}{2}$$

$$\Rightarrow K = 0.405$$

$$\therefore 3P_0 = P_0 \times e^{(0.405)t}$$

$$\Rightarrow t = 2.71 \text{ hr}$$

HW

A breeder reactor converts relatively stable uranium 238 into 239 isotope plutonium. After 15 years it is determined that 0.043% of the initial amount A_0 of population has disintegrated. Find the half life of the isotope if the rate of disintegration is proportional to the isotope.

$$t=0 \rightarrow A=A_0$$

$$t=15 \rightarrow A=0.99957 A_0$$

$$t=? \xrightarrow{\text{so that}} A = \frac{A_0}{2}$$

$$\boxed{\frac{dA}{dt} \propto A}$$

Newton's Law of Cooling/Warming: **

The rate at which the temperature of a body changes is proportional to the difference between the temperature of the body (T) and the temperature of the surrounding temperature (T_m).

Mathematically,

$$\frac{dT}{dt} \propto (T - T_m)$$

Ex-1: When a cake is removed from an oven, its temperature is measured at 300°F . 3 min later, its temperature ~~function~~ 200°F . Find the temperature function $T(t)$ where the room temperature is 70°F .

Solⁿ: we know that,

$$\frac{dT}{dt} \propto (T - T_m)$$

$$\Rightarrow \frac{dT}{dt} = K(T - T_m)$$

Given, $T = 300^\circ\text{F}$

$t = 0 \text{ min.}$

$t = 3 \text{ min.}$

$T = 200^\circ\text{F}$

$T_m = 70^\circ\text{F}$

$$\Rightarrow \int \frac{dT}{T - 70} = K \int \frac{dt}{dt}$$

$$\Rightarrow \ln(T - 70) = Kt + \ln C$$

$$\Rightarrow \ln(T - 70) - \ln C = Kt$$

$$\Rightarrow \ln\left(\frac{T - 70}{C}\right) = Kt$$

$$\Rightarrow \frac{T - 70}{C} = e^{Kt}$$

$$T - 70 = C \cdot e^{Kt}$$

$$\Rightarrow T = 70 + C e^{Kt} \quad \text{--- (i)}$$

when, $t = 0$, $T = 300^\circ\text{F}$

$$300 = 70 + C \cdot e^{K(0)} \Rightarrow C = 230$$

when, $t = 3$; $T = 200^\circ\text{F}$

$$200 = 70 + C e^{K(3)}$$

$$\Rightarrow K = \frac{\ln\left(\frac{13}{23}\right)}{3} = -0.192$$

** Linear 1st Order ODE:

of the form,

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

$$\Rightarrow \frac{dy}{dx} + \frac{a_0(x)}{a_1(x)} y = \frac{g(x)}{a_1(x)}$$

Standard form $\left\{ \frac{dy}{dx} + p(x)y = Q(x) \right\}$ — (i) where

$$p(x) = \frac{a_0(x)}{a_1(x)}$$

$$Q(x) = \frac{g(x)}{a_1(x)}$$

• Steps:

① Write a linear eqⁿ in standard form $\frac{dy}{dx} + p(x)y = Q(x)$

② Find the integrating factor, $I.F = e^{\int p(x) dx}$

③ Multiply the both sides by I.F.

The left side can be automatically written as -

$$\frac{d}{dx} [D.V \times I.F] = I.F \times Q(x)$$

④ Integrate both sides & solve for y

*

Ex:

$x \cdot \frac{dy}{dx} - 4y = x^6 \cdot e^x$. Solve the ODE

Solⁿ:

$$\frac{dy}{dx} - \frac{4}{x}y = \frac{x^5 e^x}{x}$$

$$\Rightarrow \frac{dy}{dx} + \left(-\frac{4}{x}\right)y = x^5 e^x \quad \text{--- (i)}$$

Here, $p(x) = -\frac{4}{x}$

$$\text{I.F.} = e^{\int p(x) dx} = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4} = \frac{1}{x^4}$$

$$\text{(i)} \Rightarrow \frac{1}{x^4} \left[\frac{dy}{dx} + \frac{-4}{x}y \right] = \frac{x^5 e^x}{x^4}$$

$$\Rightarrow \frac{d}{dx} \left[y \cdot \frac{1}{x^4} \right] = x \cdot e^x$$

$$\Rightarrow \int d\left(\frac{y}{x^4}\right) = \int x e^x dx$$

$$\Rightarrow \left(\frac{y}{x^4}\right) = x \int e^x dx - \int \left\{ \frac{dx}{dx} \int e^x dx \right\} dx$$

$$\Rightarrow \frac{y}{x^4} = x e^x - e^x + C$$

$$\Rightarrow y = x^5 e^x - x^4 e^x + C \cdot x^4$$

$$\therefore y = x^5 e^x - x^4 e^x + C$$

$$y = x^5 e^x - x^4 e^x + C$$

(Note: The value of C is determined by the initial condition.)

* Exact Differential Equation:

A differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be Exact D.E, if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

• Method of solution:

we know that,

$$M = \frac{\partial f}{\partial x}$$

$$\Rightarrow \int \frac{\partial f}{\partial x} = \int M \cdot dx$$

$$\Rightarrow f = \int M dx + g(y) \quad \text{--- (i)}$$

Now,

$$N = \frac{\partial f}{\partial y}$$

$$\Rightarrow \frac{\partial}{\partial y} \left[\int M dx + g(y) \right] = N$$

$$\Rightarrow \frac{\partial g}{\partial y} = N - \frac{\partial}{\partial y} \left[\int M dx \right]$$

$$\Rightarrow \int \frac{\partial g}{\partial y} = \int \left[N - \frac{\partial}{\partial y} \left(\int M dx \right) \right] dy$$

$$\Rightarrow \int \frac{\partial g}{\partial y} = \int \left[N - \frac{\partial}{\partial y} \left(\int M dx \right) \right] dy$$

$\therefore g = \dots$ (then put the value of g to (i) & we get the solⁿ.)

Ex-1: Solve the ODE:

$$2xy dx + (x^2 - 1) dy = 0 \quad \text{--- (i)}$$

Solⁿ: Here,

$$M = 2xy$$

$$N = x^2 - 1$$

$$\frac{\partial M}{\partial y} = 2x \quad ; \quad \frac{\partial N}{\partial x} = 2x \quad \left| \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \right.$$

So, eqⁿ (i) is Exactly DE

$$M = \frac{\partial f}{\partial x}$$

$$\Rightarrow \int \partial f = \int M \partial x$$

$$\Rightarrow f = \int 2xy \partial x$$

$$\therefore f = yx^2 + g(y) \quad \text{--- (ii)}$$

$$N = \frac{\partial f}{\partial y}$$

$$\Rightarrow \frac{\partial f}{\partial y} = x^2 - 1$$

$$\Rightarrow \frac{\partial}{\partial y} [yx^2 + g(y)] = x^2 - 1$$

$$\Rightarrow x^2 + \frac{\partial g}{\partial y} = x^2 - 1$$

$$\Rightarrow \frac{\partial g}{\partial y} = -1$$

$$\Rightarrow \int \partial g = \int -1 \partial y$$

$$\therefore g = -y$$

From (ii),

$$f = x^2y - y \quad \text{(Ans.)}$$

Hw

$$\textcircled{1} (3x^2y - 6x) dx + (x^3 + 2y) dy = 0$$

$$\textcircled{2} (2x + y - 1) dy + (2y - x - 5) dx = 0$$

$$\textcircled{3} (y - x) dx + (x + y) dy = 0$$

Exact

Ex: (1) $(3x^2y - 6x)dx + (x^3 + 2y)dy = 0$ — (i)

Solⁿ:

$M = 3x^2y - 6x$; $N = x^3 + 2y$

$\therefore \frac{\partial M}{\partial y} = 3x^2$; $\frac{\partial N}{\partial x} = 3x^2$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ Eq (i) is Exact

$\therefore \int M dx + \int N (\text{not containing } x) dy = 0$

$\Rightarrow \int (3x^2y - 6x) dx + \int 2y dy = 0$

$\Rightarrow 3y \cdot \frac{x^3}{3} - \frac{6x^2}{2} + 2 \cdot \frac{y^2}{2} = C$

$\Rightarrow x^3y - 3x^2 + y^2 = C$ (Ans)

$0 = y^2(x^3 + 2) + x^3(-y^2) - y^2$ (ii)

$1 - x = \frac{y^2}{x^2}$

$1 - x = \frac{y^2}{x^2} + x$

$1 = \frac{y^2}{x^2} + 2x$

$y^2 - 1 = y^2x$

$y = 1$

$$(2) (2x+y-1)dy + (2y-x-5)dx = 0 \quad \text{--- (i)}$$

$$N = 2x + y - 1$$

$$M = 2y - x - 5$$

$$\therefore \frac{\partial M}{\partial y} = 2 \quad ; \quad \frac{\partial N}{\partial x} = 2$$

$$\therefore \boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}} \quad \checkmark$$

So, (i) is Exact.

$$\therefore \int M dx + \int N (\text{Not containing } x) dy = 0$$

$$\Rightarrow \int (2y - x - 5) dx + \int (y - 1) dy = 0$$

$$\Rightarrow 2yx - \frac{x^2}{2} - 5x + \frac{y^2}{2} - y = C$$

$$(3) (y-x)dx + (x+y)dy = 0$$

$$M = y - x \quad ; \quad N = x + y$$

$$\frac{\partial M}{\partial y} = 1 \quad ; \quad \frac{\partial N}{\partial x} = 1$$

$$\therefore \boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

Exp (i) is Exact

$$\therefore \int M dx + \int N (\text{not containing } x) dy = 0$$

$$\Rightarrow \int (y-x) dx + \int y dy = 0$$

$$\Rightarrow xy - \frac{x^2}{2} + \frac{y^2}{2} = C \quad (\text{Ans.})$$

* Non-exact DE: $Mdx + Ndy = 0 \rightarrow \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

If the non-linear DE is not exact, then it can be made exact.

• Rough ✓
• Math-art ✓

Case-1: If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x alone,

then an integrating factor can be calculated as $I.F. = e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx}$

Case-2: If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M}$ is a function of y alone, then

an integrating factor can be calculated as $I.F. = e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} dy}$

then, multiply the D.E by $I.F.$, therefore we get a new exact D.E.

Note: There'll be new M & N after ^{converting} making it into exact DE.

Ex-1:

$$xy dx + (2x^2 + 3y^2 - 20) dy = 0 \quad \text{--- (i)}$$

Solⁿ: Here, $M = xy$

$$\therefore \frac{\partial M}{\partial y} = x$$

$$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Therefore, (i) is non-exact.

Now,

$$\frac{\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y}}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

Here, x, y both term exists, but we need x term alone!

$$\frac{\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y}}{-M} = \frac{x - 4x}{-xy} = \frac{3}{y} \quad \checkmark$$

Here, we needed only y term and yes, we got y term alone...

$$I.F = e^{\int \left(\frac{\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y}}{-M} \right) dy} = e^{\int \frac{3}{y} dy} = e^{3 \ln y} = e^{\ln y^3} = y^3$$

$$\therefore I.F = y^3$$

Now $I.F \times (i) \Rightarrow xy^4 dx + (2y^3x^2 + 3y^5 - 20y^3) dy = 0 \quad \text{--- (ii)}$

So, $M' = xy^4$

$$N' = 2y^3x^2 + 3y^5 - 20y^3$$

$$\therefore \frac{\partial M'}{\partial y} = 4xy^3$$

$$\therefore \frac{\partial N'}{\partial x} = 4xy^3$$

$$\therefore \frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$$

$\therefore$ Eqⁿ (ii) is exact.

Now

$$\int M' dx + \int N' (\text{not containing } x) dy = 0$$

$$\Rightarrow \int xy^4 dx + \int (3y^5 - 20y^3) dy = 0$$

$$\Rightarrow y^4 \cdot \frac{x^2}{2} + 3 \cdot \frac{y^6}{6} - 20 \cdot \frac{y^4}{4} = C$$

$$\Rightarrow \frac{x^2 y^4}{2} + \frac{y^6}{2} - 5y^4 = C$$

HW

$$1. y(1+xy) dx + x(1-xy) dy = 0$$

$$2. (x^2 + y^2 + x) dx + xy dy = 0$$

$$3. [y^3(x+1) + y] dx + (2xy+1) dy = 0$$

More Practice: Zill

Ex- 2.4 ↘

1-5, 7, 9, 10, 12,
13, 17-26

* Homogeneous DE:-

A linear n -th order ordinary differential equation of the form

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

- is said to be homogeneous if $g(x) = 0$ and non-homogeneous if $g(x) \neq 0$.

Example:- $2y'' + 3y' - 5y = 0 \longrightarrow$ Homogeneous 2nd order Linear ODE

$2y''' + 6y' + 10y = e^x \longrightarrow$ Non-homogeneous 3rd order Linear ODE

Superposition Principle - Homogeneous Equations:-

Let, $y_1, y_2, \dots, y_n$ be solutions of the homogeneous n -th order differential equation which is mentioned earlier. Then the linear combination:

$$y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$$

where the $C_i, i=1, 2, \dots, n$ are arbitrary constants, is also a solution of the equation.

* A set of two functions $f_1(x)$ and $f_2(x)$ is linearly independent when neither a function is a constant multiple of the other on the interval.

Homogeneous Linear Equations with constant co-efficients:-

We begin by considering the special case of second-order equation:-

$$ay'' + by' + cy = 0 \quad \text{--- (i)}$$

The equation has constant co-efficients a, b and c .

Since $g(x) = 0$, it's homogeneous. There is no non-linear term in the equation.

Let the trial solution of Eqⁿ (i) be of the form:-

$$y = e^{mx} \quad \text{--- (ii)}$$

Substituting (ii) $\rightarrow$ (i),

we get,

$$am^2 e^{mx} + bme^{mx} + ce^{mx} = 0$$

$$\Rightarrow am^2 + bm + c = 0 \quad \text{--- (iii)} \quad [\text{Dividing by } e^{mx}]$$

This eqⁿ is called the auxiliary equation of the differential equation (i).

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} = y$$

Solving (ii), we get,

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

1. If $b^2 - 4ac > 0$, the roots are **real** and **distinct**
2. If $b^2 - 4ac = 0$, the roots are **real** and **equal**
3. If $b^2 - 4ac < 0$, the roots are **complex conjugate**

The general solution is called **complementary solution**.

•> **Distinct Real Roots**: Under the assumption that the **auxiliary equation** has two unequal roots m_1 and m_2 , we find the general solution as -

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

•> **Repeated Real Roots**: When $m_1 = m_2$ we obtain the general solution as -

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

→ Conjugate Complex Roots: If m_1 and m_2 are complex,

then we can write -

$$m_1 = \alpha + i\beta$$

$$m_2 = \alpha - i\beta$$

Then, the form of general solution is

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

Euler's formula can be written as -

$$e^{i\theta} = \cos \theta + i \sin \theta$$

It follows from this formula that -

$$e^{i\beta x} = \cos \beta x + i \sin \beta x$$

$$e^{-i\beta x} = \cos \beta x - i \sin \beta x$$

Consequently, the general solution is:

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$$

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$y = c_1 e^{\frac{\alpha}{2}x} + c_2 e^{\frac{\alpha}{2}x}$$

Example:- Solve the following differential equation-

a) $2y'' - 5y' - 3y = 0$

b) $y'' - 10y' + 25y = 0$

c) $y'' + 4y' + 17y = 0$, $y(0) = -1$, $y'(0) = 2$

Solution: (a) Given that,

$$2y'' - 5y' - 3y = 0 \quad \text{--- (i)}$$

Let the trial solution be of the form-

$$y = e^{mx} \quad \text{--- (ii)}$$

Substituting (ii) $\rightarrow$ (i), we get,

$$2m^2 e^{mx} - 5m e^{mx} - 3e^{mx} = 0$$

$$\Rightarrow 2m^2 - 5m - 3 = 0$$

This above eqⁿ is the auxiliary equation.

Now, $2m^2 - 5m - 3 = 0$

$$\Rightarrow 2m^2 - 6m + m - 3 = 0$$

$$\Rightarrow 2m(m-3) + 1(m-3) = 0$$

$$\Rightarrow (m-3)(2m+1) = 0$$

$$\therefore m = -\frac{1}{2}, 3$$

You can solve the eqⁿ by using the formula of solving quadratic eqⁿ as well.

Therefore the general solution is-

$$y = C_1 e^{-\frac{x}{2}} + C_2 e^{3x}$$

(b) Given that,

$$y'' - 10y' + 25y = 0 \quad \text{--- (i)}$$

Let, $y = e^{mx}$ [Trial solution]
 --- (ii)

(ii) $\rightarrow$ (i)

$$m^2 e^{mx} - 10m e^{mx} + 25 e^{mx} = 0$$

$$\Rightarrow m^2 - 10m + 25 = 0$$

$$\Rightarrow (m)^2 - 2 \cdot m \cdot 5 + 5^2 = 0$$

$$\Rightarrow (m - 5)^2 = 0$$

$$\Rightarrow m = 5, 5$$

Therefore, the general solution is,

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

(C) Given that,

$$4y'' + 4y' + 17y = 0 \quad \text{--- (i)}$$

Let, the solⁿ of eqⁿ (i) be the form-

$$y = e^{mx} \quad \text{--- (ii)}$$

(ii) $\rightarrow$ (i), we get,

$$4m^2 e^{mx} + 4m e^{mx} + 17e^{mx} = 0$$

$$\Rightarrow 4m^2 + 4m + 17 = 0$$

$$\Rightarrow m = \frac{-4 \pm \sqrt{16 - 68 \cdot 4}}{2 \cdot 4}$$

$$\Rightarrow m_1 = -\frac{1}{2} + 2i$$

$$m_2 = -\frac{1}{2} - 2i$$

$$\text{Here, } \alpha = -\frac{1}{2}, \beta = 2$$

Therefore, the general solution is,

$$y = e^{-\frac{1}{2}x} (C_1 \cos 2x + C_2 \sin 2x)$$

$$\text{When, } x=0, y=-1$$

$$-1 = C_1 \cos(0) + C_2 \sin(0)$$

$$\Rightarrow C_1 = -1$$

$$y'(x) = e^{-\frac{x}{2}} (-2C_1 \sin 2x + 2C_2 \cos 2x) - \frac{1}{2} e^{-\frac{x}{2}} (C_1 \cos 2x + C_2 \sin 2x)$$

$$\Rightarrow y'(0) = 2$$

$$\Rightarrow 2C_2 - \frac{1}{2}C_1 = 2$$

$$\Rightarrow 2C_2 + \frac{1}{2} = 2$$

$$\therefore C_2 = \frac{3}{4}$$

Therefore,

$$y = e^{-\frac{x}{2}} \left(-\cos 2x + \frac{3}{4} \sin 2x \right)$$

(Ans: y)

Method of Undetermined co-efficients:-

We use the method of undetermined coefficients to solve the non-homogeneous ODE. Then the general solution has two parts - **complementary solution** and **particular solution**

i.e.,

$$y = y_c + y_p \rightarrow \begin{array}{l} \downarrow \\ \text{Complementary} \\ \text{Solution} \end{array} \quad \begin{array}{l} \rightarrow \\ \text{Particular} \\ \text{Solution} \end{array}$$

The function $g(x)$ can be constant, polynomial function, an exponential function, a sine or cosine function or finite sums and products of these functions.

Example: Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$, $y(0) = 0$

Solution: Given that,

$$y'' + 4y' - 2y = 2x^2 - 3x + 6 \quad \text{--- (i)}$$

we first solve the associated homogeneous equation

$$y'' + 4y' - 2y = 0 \quad \text{--- (ii)}$$

Let the trial solution of (ii) be of the form.

$$y = e^{mx}$$

(ii) $\rightarrow$ (i), we get,

$$m^2 e^{mx} + 4m e^{mx} - 2e^{mx} = 0$$

$$\Rightarrow m^2 + 4m - 2 = 0$$

$$\Rightarrow m = \frac{-4 \pm \sqrt{16 - 4(-2)(1)}}{2 \cdot 1}$$

$$\Rightarrow m = -2 \pm \sqrt{6}$$

Hence, the complementary solution is,

$$y = C_1 e^{(-2+\sqrt{6})x} + C_2 e^{(-2-\sqrt{6})x}$$

Because of the function $g(x)$ is a quadratic polynomial let us assume a particular solution that is also in the form of quadratic polynomial -

$$y_p = Ax^2 + Bx + C$$

We seek to determine the specific coefficients A, B, C . Substituting y and its derivatives into (i), we get,

$$2A + 8Ax + 4B + 2Ax^2 - 2Bx - 2C = 2x^2 + 3x + 6$$

Equating the co-efficients of like powers,

$$\begin{aligned} -2A &= 2 & 8A - 2B &= -3 & 2A + 4B - 2C &= 6 \\ \Rightarrow A &= -1 & \Rightarrow B &= \frac{-5}{2} & \Rightarrow C &= -9 \end{aligned}$$

Thus, a particular solution is -

$$y_p = -x^2 - \frac{5}{2}x - 9$$

Therefore, the general solution is -

$$y = C_1 e^{(-2+\sqrt{6})x} + C_2 e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

Since, $y(0) = 0$

$$\Rightarrow C_1 e^0 + C_2 e^0 - 9 = 0$$

$$\Rightarrow C_1 + C_2 = 9 \quad \text{--- (iii)}$$

and, $y'(0) = 1$

$$\Rightarrow (-2+\sqrt{6})C_1 - (2+\sqrt{6})C_2 - \frac{5}{2} = 0$$

$$\Rightarrow (-2+\sqrt{6})C_1 - (2+\sqrt{6})C_2 = \frac{5}{2} \quad \text{--- (iv)}$$

Solving, (iii), (iv), we get,

$$C_1 = 8.68, \quad C_2 = 0.315$$

$$\therefore y = 8.68 e^{(-2+\sqrt{6})x} + 0.315 e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$$

(Ans.)

Trial Particular Solution

$g(x)$	Form of y_p
--------	---------------

Any Constant	→ A
--------------	-------

$mx+c$ (linear function)	→ $Ax+B$
--------------------------	----------

mx^2+dx+f (Quadratic function)	→ Ax^2+Bx+C
----------------------------------	---------------

mx^3+nx^2+dx+f (Cubic function)	→ Ax^3+Bx^2+Cx+E
-----------------------------------	--------------------

$\sin \alpha x / \cos \alpha x$	→ $A \cos \alpha x + B \sin \alpha x$
---------------------------------	---------------------------------------

$e^{\alpha x}$	→ $A e^{\alpha x}$
----------------	--------------------

$(mx \pm c) e^{\alpha x}$	→ $(Ax+B) e^{\alpha x}$
---------------------------	-------------------------

$e^{\alpha x} \sin \beta x$	→ $A e^{\alpha x} \cos \beta x + B e^{\alpha x} \sin \beta x$
-----------------------------	---

$\alpha x^2 \sin \beta x$	→ $(Ax^2+Bx+C) \cos \beta x$ $+ (Ex^2+Fx+G) \sin \beta x$
---------------------------	--

$x e^{\alpha x} \cos \beta x$	→ $(Ax+B) e^{\alpha x} \cos \beta x$ $+ (Cx+E) e^{\alpha x} \sin \beta x$
-------------------------------	--

Example: - Solve $y'' + y = 4x + 10 \sin x$, $y(\pi) = 0$
 $y'(\pi) = 2$

Solⁿ: Given,

$$y'' + y = 4x + 10 \sin x \quad \text{--- (i)}$$

The associated homogeneous equation is,

$$y'' + y = 0$$

The complementary solⁿ is -

$$y_c = C_1 \cos x + C_2 \sin x$$

Let, the form of particular solution y_p be -

$$y_p = Ax + B + D \cos x + E \sin x$$

Substituting the particular solution y_p and its derivatives in eqⁿ (i), we get -

$$-D \cos x - E \sin x + Ax + B + D \cos x + E \sin x = 4x + 10 \sin x$$

$$\Rightarrow Ax + B = 4x + 10 \sin x$$

It is not possible to determine the values of the co-efficients from the last equation. It happens because of the repetition of $\cos x$ and $\sin x$ in the particular solution.

To avoid the repetition of terms, we set the particular solution as -

$$y_p = Ax + B + Cx \cos x + Ex \sin x$$

Then, substitution of y_p and its derivatives in eqⁿ (i) yields,

$$Ax + B - 2C \sin x + 2E \cos x = 4x + 10 \sin x$$

Solving, $A = 4$, $B = 0$, $C = -5$, $E = 0$

Therefore, the general solution is:

$$y = c_1 \cos x + c_2 \sin x + 4x - 5x \cos x$$

Firstly, $y(x) = c_1 \cos x + c_2 \sin x + 4x - 5x \cos x = 0$

$$\Rightarrow c_1 + 4 + 5 = 0$$

$$\Rightarrow c_1 = -9$$

Secondly,

$$y'(x) = -c_1 \sin x + c_2 \cos x + 4 - 5 \cos x + 5x \sin x$$

$$\Rightarrow y'(x) = -c_1 \sin(x) + c_2 \cos(x) + 4 - 5 \cos(x) + 5x \sin(x)$$

$$\Rightarrow 2 = -c_2 + 4 + 5$$

$$\Rightarrow c_2 = 7$$

Therefore, the solution to the initial value problem

is,

$$y = -9 \cos x + 7 \sin x + 4x - 5x \cos x$$

Homogeneous Linear Equations with constant co-efficients (Revisited)

Example:- Solve $y^{(4)} + y''' + y'' = 0$

Solⁿ: Given,

$$y^{(4)} + y''' + y'' = 0 \quad \text{--- (i)}$$

Let, the trial solution of eqⁿ (i) be of the form-

$$y = e^{mx} \quad \text{--- (ii)}$$

(ii) $\rightarrow$ (i), we get,

$$m^4 e^{mx} + m^3 e^{mx} + m^2 e^{mx} = 0$$

$$\Rightarrow m^4 + m^3 + m^2 = 0$$

$$\Rightarrow m^2(m^2 + m + 1) = 0$$

$$m^2 = 0$$

$$m = 0, 0$$

$$m^2 + m + 1 = 0$$

$$\therefore m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$$

The general solution is -

$$y = c_1 e^0 + c_2 x e^0 + e^{-1/2x} \left(c_3 \cos \frac{\sqrt{3}x}{2} + c_4 \sin \frac{\sqrt{3}x}{2} \right)$$

$$= c_1 + c_2 x + e^{-1/2x} \left(c_3 \cos \frac{\sqrt{3}x}{2} + c_4 \sin \frac{\sqrt{3}x}{2} \right)$$

(Ans.)

Example: Solve $16y'''' + 24y'' + 9y = 0$

Solⁿ: Given,

$$16y'''' + 24y'' + 9y = 0 \quad \text{--- (i)}$$

Let,

$$y = e^{mx} \quad \text{--- (ii)}$$

(ii) $\rightarrow$ (i)

$$16m^4 e^{mx} + 24m^2 e^{mx} + 9e^{mx} = 0$$

$$\Rightarrow 16m^4 + 24m^2 + 9 = 0$$

$$\Rightarrow 16(m^2)^2 + 24m^2 + 9 = 0$$

$$\Rightarrow m^2 = \frac{-24 \pm \sqrt{24^2 - 4 \cdot 16 \cdot 9}}{2 \cdot 16}$$

$$\Rightarrow m^2 = -\frac{3}{4}$$

$$\Rightarrow m = \pm \frac{\sqrt{3}i}{2}, \pm \frac{\sqrt{3}i}{2}$$

The general solution is,

$$y = C_1 \cos \frac{\sqrt{3}x}{2} + C_2 \sin \frac{\sqrt{3}x}{2} + C_3 x \cos \frac{\sqrt{3}x}{2} + C_4 x \sin \frac{\sqrt{3}x}{2}$$

Practice Problem:

$$\text{Ex} \rightarrow 2.4 \rightarrow (1-28)$$

$$\text{Ex} \rightarrow 4.3 \rightarrow (1-40)$$

$$\text{Ex} \rightarrow 4.4 \rightarrow (1-40)$$